



LITERATURE SURVEY REPORT

Context-Aware Explainable AI Curator for Virtual Museums using Retrieval-Augmented Generation

Submitted by

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1. Introduction

Virtual museums use AR,VR and AI to keep heritages still relevant and alive and share it with people around the world. Main feature is a virtual guide or curator powered by AI. This guide lets visitors ask questions in human language about different artifacts.

A key method that helps make these AI guides accurate is Retrieval Augmented Generation(RAG). It works by letting the AI fetch information from trusted museum documents before giving response. This keeps responses correct since the answers are based on real sources rather than just what the model remembers.

Most AI curators today act like black boxes. They give answers without showing where the information came from. They don't say how sure they are or explain why they chose that answer. That makes it hard for users to trust or understand the guidance.

This project aims to solve that issue by building a Context-Aware Explainable AI Curator. It brings together Augmented Reality, RAG and Explainable AI. Every time the system answers a question it shows the source exhibit, the document used, a confidence score, and a short explanation of how it reached the answer. The goal is to make the AI's thinking clear, transparent and trustworthy.

1.1 Importance of the Research Topic

Digital museums are popular because they are easily accessible. Digital museums can reach people who cannot access a real museum. But visitors still expect the AI guide to be more than just only giving response. Visitors want the AI guide to be trustworthy. RAG combined with XAI provides benefits:

- Increases user trust by showing the evidence behind each answer the AI guide gives.
- Reduces misinformation by forming answers on verified museum documents.
- Enhances learning by letting visitors trace an answer back to the original exhibit or catalog entry.
- Supports curators by giving transparent AI help instead of a black-box system.
- Improves engagement by providing personalized, context-aware interaction while staying credible.

1.2 Current Challenges in the Domain

- Lack of explainability-Conversational AI systems give answers without showing how they arrived at them or what evidence they used.
- Hallucinated responses-Large language models can create answers that sound right but are not correct when there is not enough real information available.
- Limited context awareness-Many systems do not pay attention to what the visitor is looking at right now what they like or what was said before so they give answers that are not useful.
- Insufficient source attribution-Chatbots usually do not say where they got the information from
- Low user trust-People might not believe the information if they do not know how sure the AI is or why it says something.
- Fragmented integration-Methods like RAG and XAI have mostly been studied on their own few systems bring both together inside an immersive AR or VR museum setting.

1.3 Objective of the Survey

This survey reviews recent research (2021–2026) on AI-assisted virtual/AR/VR museum guides, Retrieval-Augmented Generation, and explainable conversational systems, in order to

- (a) understand existing methods and their results,
- (b) identify common limitations
- (c) establish the research gap that the proposed Context-Aware Explainable AI Curator is designed to fill.

2. Literature Survey Table

S.No	Authors & Year	Paper Title	Methodology / Algorithm	Dataset / Tools	Key Results	Limitations / Research Gap
1	de Oliveira et al., 2025	Immersive Virtual Museums with Spatially-Aware Retrieval-Augmented Generation (ACM IMX '25)	Combines eye-tracking (gaze/visual attention) with RAG and an LLM to ground VR museum guidance in what the visitor is looking at	VR museum environment, exhibit metadata, eye-tracking gaze traces, vector retrieval	Gaze-guided retrieval gave more contextually relevant answers than a generic LLM guide; improved personalization	Prototype-scale only; no confidence score, no source citation, needs eye-tracking hardware
2	Kerle-Malcharek et al., 2026	Design Space and Implementation of RAG-Based Avatars for Virtual Archaeology (arXiv)	Design framework (realm, personality, interaction) plus a RAG-based conversational VR avatar for an archaeological site	Curated archaeological text corpus, expert QA benchmark, VR workload questionnaires	Avatar supported topical, on-subject conversations and achieved low reported workload among participants. The study evaluated the system using 15 participants and expert-created QA pairs.	Small-scale evaluation; limited treatment of transparency and uncertainty, particularly regarding the authenticity/uncertainty of reconstructed archaeological content.
3	Filho et al., 2026	Image-to-3D for Cultural Heritage: Voice Interaction and Retrieval-Augmented Guidance in VR (IEEE AIXVR)	Reconstructs a 3D artifact from a single photograph and pairs it with a voice-controlled RAG assistant	Exhibition photographs, voice commands, 3D-generation pipeline	Demonstrates a pipeline combining single-image 3D reconstruction with voice interaction and retrieval-augmented guidance for virtual cultural-heritage exhibits	3D quality lower than multi-view reconstruction; no explainability or citations
4	Ariya et al., 2025	Enhancing Textile Heritage Engagement through Generative AI-Based Virtual Assistants in VR Museums (Computers & Education: X Reality)	Mixed-methods study of a multilingual, voice-based AI docent inside a VR textile-heritage reconstruction	Textile heritage dataset; 25 university-student participants	High user satisfaction; visual quality was rated 4.40/5 and ease of interaction 4.20/5 . Response latency (3.36/5) and voice clarity (3.62/5) were areas for improvement.	Small sample size; no traceability, explainability, or confidence estimation
5	Çakmak & Çekmiş, 2026	AI-Directed Procedural Content Generation for Personalized XR in Industrial-Heritage Museums (Frontiers in VR)	Rule-based 'AI Director' classifies visitor movement/gesture and adapts the museum flow using procedural content generation	XR tracking data, controller pose, Catmull–Rom splines	Successfully classified visitor behaviour and personalized the exhibit sequence	Uses a rule-based AI Director rather than an LLM/RAG conversational architecture; therefore it does not address retrieval-grounded natural-language question answering or explainable conversational responses.
6	Konstantakis & Iakovaki, 2026	ARIA: An AI-Supported Adaptive Augmented Reality Framework for Cultural Heritage (Information, MDPI)	Conceptual framework combining multimodal context, affect recognition, and RAG for adaptive AR storytelling	CIDOC-CRM heritage-metadata standards	Argues that RAG plus affect-awareness can improve adaptive storytelling and trust	Conceptual only — no implementation, benchmark, or user study
7	Kostadimas et al., 2025	Exploiting VR, AIoT and Semantics Towards an Adaptive Virtual Museum (IEEE SMAP)	Integrates VR, AIoT sensors, semantic-web ontologies, and an LLM guide for visitor profiling	HMD behavioural traces, digital-twin model	Enables intelligent visitor profiling and semantic route optimisation	Proposed adaptive virtual-museum architecture combining VR, AIoT and semantic technologies; further implementation and empirical validation are required to establish its effectiveness.
8	Kumari et al., 2025	Virtual Art Spaces and Machine Learning Curation (ShodhKosh: Journal of Visual and Performing Arts)	Systematic literature review of ML methods (classification, style transfer, similarity learning) used for exhibition curation	Survey of published ML-curation literature	ML improves artifact classification and thematic exhibition organisation	Review paper only; no implementation and no RAG/XAI discussion
9	Li, 2024	Opportunities and Challenges of Artificial Intelligence Enabling Museum Building (Applied	RBF interpolation plus GC-GAN restoration and texture mapping to	Dunhuang mural point-cloud data; SUS and UEQ usability scores	High restoration accuracy (SSIM 0.83, PSNR 26.36 dB) and good usability (SUS 73.9)	Focused only on visual restoration; no conversational AI or explainability

		Mathematics and Nonlinear Sciences)	repair damaged 3D point clouds before VR display			
10	Ly et al., 2025	Museum in the Classroom: Engaging Students with Augmented Reality Museum Artifacts and Generative AI (CHI EA '25)	AI chatbot integrated with curated artifact libraries inside an iPad-based classroom AR application	Curated artifact collection, iPad AR app	Brings museum-style exploration into the classroom without a physical visit	Classroom-focused; no VR, no RAG source citation, no explainability
11	Zhao et al., 2024	Retrieval-Augmented Generation for AI-Generated Content: A Survey (arXiv)	Comprehensive taxonomy of RAG architectures (dense/sparse/hybrid retrieval, retriever-generator coupling) and evaluation methods	Cross-domain RAG benchmark datasets	RAG measurably reduces hallucination and reliance on outdated model knowledge	General-purpose survey; not specific to museums, AR, or VR
12	Zhou et al., 2022	Meta-Analysis of VR/AR Effectiveness in Museum Learning (Educational Research Review)	Systematic literature review of 51 relevant studies, with 17 studies included in the meta-analysis of AR/VR effects on museum learning.	51 relevant studies identified; 17 studies included in the meta-analysis.	VR/AR showed significant positive effects on academic achievement (ES = 0.45) and learner perceptions (ES = 0.59).	Predates LLMs and RAG; does not discuss AI explainability
13	Zidianakis et al., 2021	The Invisible Museum: A Digital Heritage Authoring Platform (Electronics, MDPI)	Authoring platform using CIDOC-CRM, Europeana Data Model, and deep learning for semantic heritage management	Digitised artifacts and curator narratives	Enables standards-based creation of semantically rich virtual exhibitions	Focused on authoring/provenance, not conversational AI or question answering
14	Cuong Nguyen, 2026	The collaboration of artificial intelligence (AI) and museums: possible or impossible?	Qualitative discussion and case-based analysis of AI applications, opportunities, and challenges in museums.	Institutional case studies (e.g., British Museum, Louvre)	Identifies incomplete/inconsistent heritage data as a major obstacle to reliable AI answers	No technical architecture or implemented system

3. Research Gap

Table 2 summarises how the limitations observed across the reviewed literature translate into specific gaps, and how the proposed Context-Aware Explainable AI Curator is designed to close each one.

Table 2: Research Gap Summary

Research Gap	Limiation in Existing Studies	How the Proposed Work Addresses it
Lack of explainability	Most AI museum guides answer without explaining their reasoning or the evidence used.	Integrates Explainable AI (XAI) to give a clear "why this answer" explanation with every response.
Absence of source attribution	Systems rarely identify the exhibit, catalogue entry, or document an answer came from.	Displays the exact source exhibit and retrieved document backing each answer.
No confidence estimation	Users cannot judge how reliable an AI-generated answer is.	Computes and shows a confidence score based on retrieval relevance and generation quality.
Limited context awareness	Most systems use only a single contextual signal (gaze, movement, or voice alone).	Combines exhibit location, user interaction, and conversation history to personalise responses.
Fragmented RAG + XAI integration	RAG and XAI are usually studied as separate research threads.	Builds a single framework that unifies context-aware RAG with explainable output.
Prototype / conceptual systems only	Several works are conceptual frameworks or small prototypes with limited validation.	Implements and evaluates a functional prototype in a working virtual museum environment.
Limited practical evaluation	Many studies rely on small datasets or brief qualitative user studies.	Plans evaluation on realistic museum datasets using accuracy, trust, explainability, and satisfaction metrics.

4. Connection to the Proposed Project

The papers reviewed above map directly onto the three pillars of the proposed project which are immersive presentation (AR/VR), grounded retrieval (RAG) and transparency (XAI)

- Immersive delivery: Papers [1] [3] [4] [5] [7] [9] [10] and [12] demonstrate that AR/VR boosts engagement and learning in museum settings. This finding supports the decision to use an AR-based interface for the proposed curator.
- Grounded retrieval: Papers [1] [2] [3] [6] [7] and [11] show that RAG is a method to lower hallucination by anchoring an LLM's answer in verified documents. The same retrieval strategy sentence-transformer embeddings and vector search is used in the proposed methodology.
- Personalisation and context: Paper [1] uses gaze, paper [5] uses movement or gesture, and paper [7] uses sensor or semantic profiling. Each of these uses one contextual signal. The proposed system expands on this by linking exhibit location, interaction history and conversation context together.
- Transparency and trust: None of the fourteen reviewed works combine source attribution, a confidence score and an explicit reasoning explanation in a single system. This gap is what the proposed Context-Aware Explainable AI Curator addresses, within an AR virtual-museum environment powered by a FastAPI/RAG/XAI pipeline.

5. Literature Survey Summary

5.1 Existing Approaches and Common Methods/Algorithms Used

The fourteen papers reviewed for this project fall into a few clear technical categories. Understanding these categories helps to explain exactly where the proposed system fits in.

The first category is a large language model with Retrieval-Augmented Generation (RAG). This involves the system first retrieving relevant text from a curated knowledge base and passing that text to the LLM so that its answer is grounded in real, verifiable content, rather than just generating an answer based on whatever the LLM learned during training. de Oliveira et al. [1] apply this idea inside a VR museum by using eye-tracking to decide what to retrieve — the system watches what the visitor is looking at and pulls information about that specific exhibit. Kerle-Malcharek et al. [2] employ the same RAG idea but they wrap it in a conversational avatar with a defined personality making the interaction feel more interactive. Filho et al. [3] take it a step further by adding voice input and creating a 3D model of the exhibit from a single photo so the entire experience -seeing, asking and hearing is done within one pipeline. Zhao et al. [11], is not museum-specific but is important because it lays out the technical foundation upon which all of these systems rely: it describes the different types of retrieval (dense, sparse, or a hybrid of both) and how the retrieved text is fed to the generator

The second type is context-driven personalisation. This is where the system alters its behaviour based on something it observes about the visitor. All the papers in this category use one input at a time. de Oliveira et al. [1] use gaze. Çakmak and Çekmiş [5] use body movement and hand gestures that are classified by a rule-based AI Director which decides what content will be shown next. Kostadimas et al. [7] use data from IoT sensors to build a profile of the visitor and suggest a route through the museum. None of these three systems combine more than one of these signals together.

Third we have multilingual and voice interaction. Filho et al. [3] use voice as the input method to query their RAG system, and Ariya et al. [4] built and tested a multilingual, voice-based AI to allow visitors who do not speak the museum's default language to ask questions naturally. Voice interfaces are already seen as a practical way to make AI museum guides more accessible.

The fourth category covers reconstruction and 3D restoration. This category is about creating or fixing content itself not answering questions about it. Filho et al. [3] reconstruct an object from a single 2D photograph which is a fast method that produces lower-quality results than methods using multiple photographs of the same object.

The fifth category is about how data is structured in a meaningful way using semantics and ontologies. This is less about the AI model itself and more about how museum data is organised so it can be searched and used reliably. Zidianakis et al. [13] and more generally, Konstantakis and Iakovaki [6] both use CIDOC-CRM, a data model used in the museum and heritage field to describe artifacts, events and relationships in a consistent way. Zidianakis et al. [13] also use the Europeana Data Model (EDM) and deep learning to create a platform for making exhibitions that follow these standards. This category is important for the proposed project because an organised knowledge base is what makes retrieval in RAG actually dependable.

Finally there are papers that are reviews, surveys or meta-analyses instead of working systems. Kumari et al. [8] look at machine-learning methods used for exhibition curation. Zhao et al. [11] cover RAG techniques in general. Zhou et al. [12] conduct an analytic review of 65 studies on VR/AR in museum learning. Nguyen [14] discusses how real museums such as the British Museum and the Louvre use AI. These papers do not create systems but they show that the techniques used (RAG, ML curation and immersive experiences) are already well studied and proven, giving the proposed project a base to build from rather than an untested idea.

Together, this survey shows that RAG-based retrieval, single-signal personalisation, voice interaction, 3D reconstruction and semantic data structuring are all mature and tested on their own. No single paper reviewed here combines RAG-based retrieval with an explanation-and-confidence layer for the visitor. That specific mix is what the proposed project aims to build.

5.2 Important Findings from Previous Studies

De Oliveira et al [1] found that when the retrieval was guided by the visitor's gaze, the responses generated were perceived as more relevant to the situation than those generated by a regular LLM guide that did not use this kind of guidance – knowing what the visitor is looking at genuinely improves the answer not just its personalization. Kerle-Malcharek et al. [2] tested their RAG-based avatar against a question-answering set made by experts and found that it stayed accurate and on-topic when discussing an archaeological site, while users reported below-average mental effort on a standard VR workload test - the system was both accurate and easy to use. Zhao et al.'s larger study [11] states that RAG helps reduce wrong information and reliance on outdated knowledge compared with a regular LLM that has no retrieval step.

On the user experience side, Ariya et al. [4] tested their VR guide with 25 university students and recorded satisfaction scores of 4.40 out of 5 for visual quality and 4.20 out of 5 for overall experience — high scores for a system that is still a prototype, showing that visitors respond well to AI guides designed with real users in mind. Çakmak and Çekmiş [5] found that their rule-based AI system could distinguish visitor behaviours by observing movement and gestures, and can change how museum content was displayed showing that even non-LLM AI can meaningfully change the visitor experience.

Zhou et al.'s review of 65 studies [12] is an important reference that predates LLMs and RAG, but still found VR and AR to improve the learning, motivation and engagement in museums which shows that the project is designed as an AR application rather than a regular website or app as the format itself helps people learn better even without AI. Nguyen's study [14] found that one of the main obstacles to AI working well in real museums is not the AI model itself but the lack of good, well-organised information about the items underscoring the importance of building a well-organised knowledge base as part of this project not just a good language model.

5.3 Limitations of Existing Work and Research Gaps Identified

While each of these papers makes a genuine contribution to the field, reading them together reveals a consistent set of gaps that recur across the literature, regardless of the specific technique or system architecture adopted.

The first and the most obvious gap is an absence of explainability. The systems mentioned [1], [3], [4], [5], [10] provide visitors with answers or adapt their experience; nevertheless, they do not give explanations about how and why a certain answer or adaptation has been created. For instance, a visitor who uses the system by de Oliveira et al. [1] gets a gaze-grounded answer; however, he or she cannot get any idea of what document, passage or fact it has been based on. In the context of museums, such a deficiency is especially critical because visitors may be interested in finding out where the information came from and not just getting the system-generated answer.

The second gap is an absence of source attribution in the context of explainability. It concerns even such systems as [1], [2], [3] that apply the technique of RAG. The essence of RAG lies in the grounding of generated answers in relevant documents or sources of knowledge; when the system does not indicate the document or the source used, such grounding becomes invisible to the visitor.

A third, related gap is the complete absence of confidence estimation. None of the fourteen papers reviewed here — including the RAG-based systems in [1], [2], and [3] — provides visitors with any indication of how confident the system is in a particular answer. This is significant because, as Zhao et al. [11] explicitly point out, hallucination remains a potential problem even when RAG is used. Without some form of confidence estimation, a visitor has no way to distinguish between an answer the system considers highly reliable and one for which the underlying evidence is weak or uncertain.

The fourth gap concerns the limited use of contextual information for personalisation. Where personalisation is incorporated, it is generally based on a single contextual signal: [1] relies on gaze, [5] considers movement and gesture, while [7] uses sensor- and ontology-based visitor profiling. Although these approaches demonstrate the value of context-aware adaptation, none combines multiple sources of information — such as the exhibit being viewed, the visitor's current question, and previous interactions — into a unified representation of the visitor's situation.

The fifth gap concerns the limited scale and depth of evaluation, even for systems that have been implemented. Existing studies are often evaluated using small participant groups or short, predominantly qualitative experiments. For example, [4] evaluates its system with only 25 participants, while [2] relies on a relatively small expert-created benchmark. Such evaluations are useful for demonstrating feasibility but provide limited evidence about performance in a real museum environment with larger, more diverse visitor populations.

The sixth gap, and arguably the most important, is the separation between RAG and Explainable AI (XAI) in existing research. The reviewed literature largely treats these as two separate research problems: papers such as [1], [2], [3], and [11] focus primarily on retrieval, grounding, and answer generation, while other studies focus on personalisation and content adaptation [5], [7], or on underlying data structures and semantic representations [6], [8], [13]. None of the fourteen papers reviewed here integrates RAG-based retrieval with explainable, trust-building output within a single system specifically designed for an immersive AR/VR museum environment.

Taken together, these gaps point to a broader limitation in the current state of research: existing systems can retrieve information, generate answers, adapt content or respond to aspects of visitor context but they generally do not give the visitor a transparent understanding of what information was used where it came from how confident the system is and why the response was appropriate for that particular visitor and situation. This combination of limitations defines the central research gap addressed by the proposed project: there is a need for a unified, implemented, and empirically evaluated immersive AR/VR museum system that combines RAG-based retrieval, explainable AI, source attribution, confidence estimation, and multi-source visitor context to produce answers that visitors can understand, verify, and trust.

5.4 How the Proposed Project Addresses These Gaps

The proposed Context-Aware Explainable AI Curator is built to tackle the six gaps identified above, not to improve just one existing system. Every part of the design responds to a problem found in past research, bringing retrieval, personalisation, explainability, transparency and real-world testing into one experience that visitors can use while exploring a museum.

To fix the lack of explainability noted in [1], [3], [4], [5] and [10], the system includes an Explainable AI (XAI) layer that adds a plain-language explanation with every answer, stating why the answer was given and what information or context led to it answering not just "what is the answer?" but "why is this the answer?"

To fix the source-attribution problem still seen even in RAG-based systems like [1], [2] and [10], the system will show where each answer came from naming the exhibit and pointing to the exact document or catalogue entry that supports the response — making the retrieval process visible so visitors can check it against the source.

To address the lack of confidence estimation mentioned in [1] and [2], the system will display a numeric confidence score for each answer, derived from factors such as how well the retrieved material matches the question and how sure the model is in its output — helping visitors gauge trustworthiness and spot when an answer might need checking, given the hallucination risk Zhao et al. [11] highlight.

To solve the single-signal personalisation issue described in [1], [5] and [7], the system uses more than one type of visitor data — combining the exhibit being viewed, the question being asked, and the history of interactions — to build a richer view of what the visitor is doing and what information they might need.

To fix the gap where RAG and XAI are treated separately, as seen in [1], [2], [3], [5], [6], [7], [8], [11] and [13], the project brings both capabilities together in one complete pipeline. The architecture uses a Gemini-based Large Language Model (LLM) to generate answers, sentence-transformer embeddings and ChromaDB for retrieving knowledge, and an explanation-and-confidence-scoring layer to make the process clearer to the visitor — all running through an AR-based mobile front end supported by a FastAPI backend service. Explainability is built into the core of the system alongside retrieval and generation, not added afterward.

Finally, to address the implementation and evaluation gaps found in studies such as [2], [4], [6] and [7], the project aims to produce a working prototype, not just a theoretical idea, built using museum data and tested against real performance measures including response accuracy, retrieval precision, response time and user satisfaction — providing evidence of whether combining RAG, XAI, contextual personalisation, source attribution and confidence scores gives real benefits in a museum setting.

In total, the proposed system responds directly to the problems identified in the research. Rather than focusing on just retrieval or explainability, the Context-Aware Explainable AI Curator links all these features into one visitor-facing system: answers become contextually relevant, grounded in checkable sources, explained in plain language, and paired with a confidence level — built and tested in a realistic museum environment so it demonstrates not just what could work, but what actually does.

6. References

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